



and mechanical drawings. Documents are also required for any special instructions for components assembly, cable harness and integration, assembly sequence, acceptance procedures for individual components, materials, assemblies/sub-modules and total unit.

Prototype and qualification testing

Build a prototype and perform testing as per the documented procedure. Initially, prove functional performance under ambient conditions. Later, prove performance under higher stresses than required during normal operation—a process called qualification testing.

Qualification testing establishes design margins above required operating stresses for a unit. A design having a higher margin will help reduce rejections in production and reliability for a longer duration in the field. Analyse any failure to meet any requirement during this phase. Modify the design to meet the requirement. Modification could be required either in the circuit or packaging.

Critical design review

Get the design reviewed again by experts in relevant fields with respect to prototype qualification test results. Relevant modifications need to be incorporated in design documents.

Pilot production and acceptance testing

Now go for production of ten to fifteen units with qualification-test-cleared design. Component values are spread

randomly among the units within tolerance limits. Complete performance testing as per the documented procedure at application stresses. This process is called acceptance testing.

With all the design steps given earlier followed, it is much less likely that any design problem will occur. Manufacturing problems are, however, likely to surface. Identify performance problems at this stage as manufacturing or design. Manufacturing problems can be solved by repair or rework. If any design problem surfaces, modify the design and confirm solution. Modify design documents accordingly. Major design modification might require qualification testing again.

Production and quality control

The required level of quality and reliability is built into the product during design stage. It can be maintained in production only if correct materials and components are used, and correct assembly and manufacturing procedures followed as per design documents and drawings.

It is the responsibility of the quality control department to ensure that the design is translated correctly into a product during production by using right materials, components, assembly and manufacturing procedures as per design documents and drawings.

Failure reporting and corrective action

Any failure or non-conformance occurring during production quality control

and testing is to be analysed by a failure analysis board to find out the root cause.

The cause needs to be traced to the manufacturing/assembly process or design. Required improvements are recommended and implemented to avoid recurrence of such failures.

Accelerated stress screening

A product is likely to fail in its initial stages of operation after

it is delivered to the customer. This can happen if some manufacturing defects escaped industry quality control and testing procedures, or the unit was not subjected to testing under required operational stresses. These defects remain hidden in the product delivered to customers. Hidden defects precipitate under unit operational stresses at customer site for some time and lead to failure.

This situation can be avoided by testing the product at stresses higher than operational requirement but within design limits during manufacturing. Higher stresses will precipitate hidden defects in a shorter time than required under normal operational stresses at customer site.

Failed units can be reworked to remove manufacturing defects and delivered to the customer. This reduces the chance of the unit failing at the customer's site in early stages of operation. This method of testing is called accelerated stress screening.

Reliability aspects

It is possible to estimate and predict the reliability of a design based on quality grades of components used and operational stresses. It is also possible to demonstrate reliability achieved by building certain number of units with the same design and using statistical methods. Reliability can also be estimated from field failure data. ●

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